

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460398.7089 +/- 0.0055 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.166 +/- 0.049
Transit depth (Rp/Rs)^2	2.74 +/- 1.61 [%]
Orbital Inclination (inc)	87.41 +/- 2.95
Airmass coefficient 1 (a1)	0.63 +/- 0.31
Airmass coefficient 2 (a2)	0.31 +/- 0.42
Scatter in the residuals of the lightcurve fit is	50.74 %
Best Comparison Star	None
Optimal Aperture	4.61
Optimal Annulus	12.07
Transit Duration (day)	0.0917 +/- 0.01

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.166 ± 0.049 vs 0.1368 (deviation 0.52σ)
Duration fit vs published	0.0917 d vs 0.0758 d (deviation 1.38σ)
Mid-time O-C	8.1 min
Depth SNR	2.9
Residual scatter	50.74 %

Light curve

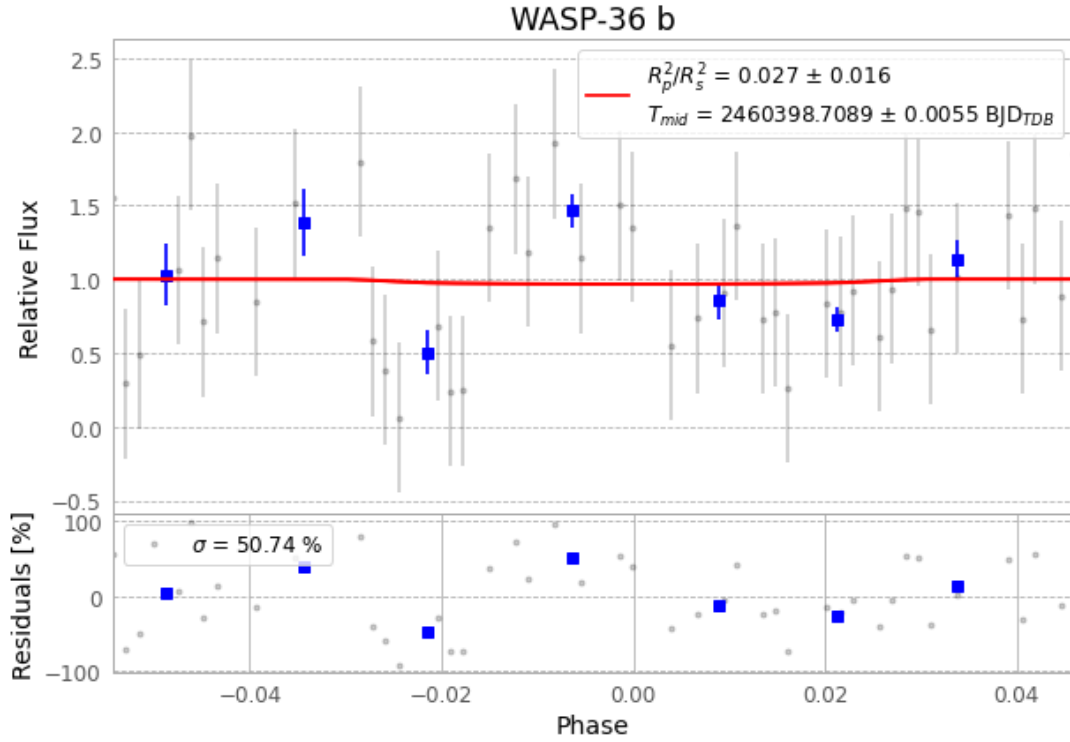


Figure 1 — FinalLightCurve_WASP-36 b_2024-03-28.png (SHA-256 9a87dcc5b4aea948...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [406, 204], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9bebae181e52905a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

77 FITS frames (51 MB), WASP-36240329025415.FITS → WASP-36240329064215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-240329.log (2f602dfbf35b...), FinalLightCurve_WASP-36 b_2024-03-28.png (9a87dcc5b4ae...), FinalLightCurve_WASP-36 b_2024-03-28.csv (d198347e3e6b...)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 07:28Z.